

Module 4: Apriori & FP–Growth Pattern Mining

Support/Confidence, Apriori Join & Prune, FP–Tree Mining, Lift

Module IV: Apriori & FP–Growth Pattern Mining — Topics Covered

1. Market Basket Analysis: Support & Confidence
2. The Apriori Downward Closure Property
3. Apriori Candidate Generation: Join & Prune
4. FP–Tree Construction & Recursive Mining
5. Pattern Correlation: Lift & Kulczynski

1. Support, Confidence & Lift Formulations

$$\text{Support}(A \implies B) = P(A \cup B) = \frac{\text{count}(A \cup B)}{|D|}$$

$$\text{Confidence}(A \implies B) = P(B | A) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

$$\text{Lift}(A, B) = \frac{P(A \cup B)}{P(A)P(B)}$$

BIT Mesra Exam Question (10 Marks)

Problem: Explain the Apriori property and demonstrate how the prune step eliminates candidate itemsets without scanning the database.

Solution: *Apriori Property:* All non-empty subsets of a frequent itemset must also be frequent. If any $(k - 1)$ -subset of candidate C_k is not present in frequent list L_{k-1} , C_k cannot be frequent and is immediately pruned without costly database I/O.